Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Integrating Hybrid Asymmetric Interfaces and Situated Reasoning for Next-Generation Smart Factories			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (), Next-gen Nuclear Power (), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Control and Intelligence Technologies	Digital Content	Virtual Reality
Concept of Proposed Technology	This project introduces a **Hybrid Asymmetric Intelligence Platform** designed to resolve the "information asymmetry" that bottlenecks efficiency in high-complexity manufacturing sites (e.g., Semiconductor Fabs and Smart Shipyards). By bridging the cognitive gap between digital data and physical reality, we propose a fundamental architectural shift in industrial digitalization. Core Technological Pillars are: 1. ***Scale-Adaptive Visualization Engine***: A dynamic rendering core that eliminates data silos. It employs a Hierarchical Level-of-Detail (HLOD) architecture to seamlessly transition			

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	between macroscopic logistics flows (e.g., entire dry docks via point cloud streaming) and microscopic precision tasks (e.g., sub-millimeter component CAD) without latency. 2. ***Asymmetric Collaborative Interface***: Recognizes that industrial work is inherently asymmetric. We replace simple screen mirroring with "Cross-Reality Translation," allowing a remote expert (on PC/VR) to guide a local technician (on AR/Mobile) via synchronized spatial anchors, bridging the hardware and skill gap. 3. ***AI-Driven Situated Analytics***: Moves beyond passive dashboards to "Actionable Intelligence." The system embeds abstract data directly into the physical context using computer vision (YOLO/VLM) to detect anomalies (e.g., hull cracks, wafer misalignment) and overlay remedial guidance in real-time, reducing cognitive load. 4. ***Sim-to-Real with Human-in-the-Loop***: A risk-free "AR-in-the-Loop" simulation framework. It allows human workers to interact with virtual robots projected into the real workspace, generating crucial safety data to train robot navigation policies before physical deployment.
Rationale for Joint R&D Project	The collaboration between Georgia Tech (GT) and the Korean SME is structurally necessary to bridge the "TRL Gap." Georgia Tech provides world-class expertise in the theoretical and algorithmic foundations of Human-Computer Interaction (HCI) and Immersive Analytics. GT supplies the "interaction logic": the proprietary algorithms, designs, and implementations that minimize cognitive load and synchronize users across heterogeneous devices. Korean SME provides the critical hardware integration (explosion-proof tablets, cleanroom-certified smart glasses) and acts as the gateway to industrial testbeds. Their established relationships with major conglomerates (e.g., HD Hyundai, Samsung Electronics) are essential for securing pilot sites and ensuring the technology is "ruggedized" for real-world deployment. Strategic Impact on Key Industries: ● Shipbuilding (The Labor Crisis): Addresses the 45,000-worker deficit by enabling "Technological Upskilling." Retiring masters can remotely supervise and train novice workers via AR, ensuring knowledge transfer and operational continuity. ● Semiconductors (The Yield Imperative): Reduces expensive equipment downtime via "Micro-Scale Digital Twins." Remote experts can diagnose angstrom-level faults through the eyes of local technicians, drastically cutting Mean Time To Repair (MTTR) and contamination risks.
Key Words	1. Virtual and Augmented Reality 2. Digital Twin 3. Smart Manufacturing 4. Digital Collaboration 5. AR Robot Simulation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The overarching objective is to develop and validate a Unified XR Digital Twin Platform that bridges the scale gap between semiconductor (micro) and shipbuilding (macro) environments. Specific Technical Objectives: 1. ***Scale-Adaptive Rendering Core***: Engineer an engine capable of dynamic switching between massive point-cloud rendering and high-fidelity parametric CAD while maintaining >60 FPS on mobile XR devices. 2. ***Asymmetric Sync Protocol***: distinct from video streaming, this low-latency architecture (WebRTC/WebSocket) synchronizes interaction states and viewports to support seamless remote expert guidance. 3. ***AI-Driven Situated Analytics***: Deploy computer vision models (VLM/YOLO) to recognize industrial objects and overlay spatially registered data (e.g., stress heatmaps) with high precision. 4. ***Human-in-the-Loop Simulation***: Capture interaction data between AR users and virtual robots to train robust Situated Reasoning policies for safe autonomous navigation.			
R&D Target Specifications	1. Motion-to-Photon Latency (< 20 ms): Critical threshold to ensure user comfort and prevent motion sickness in immersive environments. 2. End-to-End Sync Latency (< 100 ms): Maximum acceptable delay to maintain a "shared mental model" during real-time remote collaboration. 3. AR Registration Accuracy: ○ Micro-mode: < 1 mm (Essential for semiconductor circuitry overlay). ○ Macro-mode: < 2 cm (Sufficient for shipbuilding block alignment). 4. Object Recognition Rate (> 95%): Required reliability to ensure user trust in Situated Analytics. 5. Scalability: Support for > 10 concurrent asymmetric users (e.g., one Expert guiding multiple workers).			
R&D Plan	● Year 1: Foundation & Architecture (Simulation Phase) ○ Q1-Q2: Contextual Inquiry & Architecture. Conduct workflow mapping at partner sites to identify pain points. Define network topology and security protocols (IP protection). ○ Q3-Q4: Core Engine Development. GT develops the Asymmetric Sync Protocol (state synchronization) and Scale-Adaptive Rendering (LOD algorithms). SME builds the Synthetic Data Generator to create training datasets from existing CAD assets. ● Year 2: Integration & Prototyping (Lab Phase) ○ Q1-Q2: Hardware Integration. SME integrates software into industrial devices (e.g., ruggedized tablets, cleanroom AR glasses) and implements sensor fusion (IMU + Visual Odometry) for stable tracking. ○ Q3: Situated Analytics UI. Develop "World-Fixed" labels (shipbuilding) and "Object-Fixed" labels (semiconductor) with occlusion management. ○ Q4: Lab Trials. Conduct controlled user studies (GT & SME) to measure Time-to-Completion and Cognitive Load (NASA-TLX) in collaborative tasks. ● Year 3: Validation & Commercialization (Pilot Phase) ○ Q1-Q2: Field Deployment. Deploy pilots: (1) AR Block Inspection (Shipyards), (2) Remote Maintenance Support (Fabs), (3) AR-in-the-Loop Robot Training. ○ Q3: Refinement. Analyze longitudinal pilot data to refine UX and reduce long-term fatigue. ○ Q4: Commercialization. Finalize ISO certification and launch market strategy.			

4. R&R (Roles and Responsibilities)

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Expected R&R of Research Institute	Core Innovation & Algorithms: ● Visualization Design: Design intuitive "Situated Analytics" techniques to ensure data overlays are unobtrusive and cognitively efficient. ● Sync Architecture: Develop the algorithms solving "referential ambiguity," allowing seamless collaboration across VR, AR, PC, and Tablet. ● Human Factors: Conduct rigorous user studies to validate usability, training efficacy, and cognitive load reduction. ● AI Development: Build Computer Vision pipelines for object recognition and "AR-in-the-Loop" simulation frameworks to train robot policies.
Expected R&R of Korean SME Partner	Realization & Industrialization: ● Hardware Ruggedization: Adapt and optimize software for specific industrial hardware (anti-explosion, cleanroom-certified). ● Testbed Access: Leverage domestic networks to secure pilot sites at major conglomerates (Hyundai, Samsung, etc.). ● System Integration: Develop middleware to bridge the Digital Twin with legacy industrial systems (ERP/MES). ● Commercialization: Lead business modeling, ISO certification, and export strategy development.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The primary output of this project is a unified ***Cross-Scale Engine***, a first-of-its-kind digital twin platform capable of handling the extreme data dichotomy between meter-scale shipbuilding blocks and micron-scale semiconductor architectures. A critical technological breakthrough will be the validation of a robust ***Asymmetric Collaboration Protocol***, designed to solve the "referential ambiguity" that plagues current remote support tools. By synchronizing the viewports of experts and workers using different devices (e.g., VR, AR, PC, and tablet) via a low-latency web architecture, this protocol ensures a "shared mental model" even under industrial network constraints, allowing for precise remote guidance. Furthermore, the initiative establishes a continuous safety improvement cycle. This iterative methodology harvests human interaction data from AR trials to train autonomous agents, allowing them to learn "social common sense" and achieve zero-shot deployment with less than 5% of the real-world fine-tuning time traditionally required. Economically, the platform directly addresses existential threats to Korea's flagship industries by modernizing workforce capabilities through asymmetric interaction. In the shipbuilding sector, the platform's "Asymmetric Training" capabilities allow retiring master craftsmen to remotely guide novice workers via AR, directly mitigating the 45,000-worker labor shortage. This "Just-in-Time" guidance is projected to reduce training time by 50% and inspection time by 30%. Simultaneously, for the semiconductor sector, the technology enhances yield by enabling instant remote diagnostics; the asymmetric interface allows global experts to virtually "teleport" into cleanrooms to guide local technicians with millimeter precision. By reducing Mean Time To Repair (MTTR) in fabs where downtime costs exceed $100,000 per hour, the platform is poised to generate millions in annual savings while increasing Automated Material Handling System (AMHS) efficiency by an estimated 15%. On a global scale, this collaboration positions the participating Korean SME as a leader in "Industrial Metaverse (spatial computing)" solutions, moving beyond simple 3D visualization to complex, AI-driven collaborative workflows. By validating these technologies in Korea's world-leading testbeds, the project creates significant export potential to US and European markets currently seeking efficiency tools for manufacturing re-shoring. The project also aims to establish new industry standards for "Socially Compliant Robotics" by formalizing STL-based safety verification, ensuring that the future of industrial automation is not only efficient but mathematically verified to be safe.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization